

Practice problems #1

1. For the matrix $A = \begin{bmatrix} -6 & 12 \\ -3 & 6 \end{bmatrix}$, let x be the vector $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$. Is $x \in \text{Nul } A$? Why or why not? Is $x \in \text{Col } A$? Why or why not? Is x an eigenvector of A ? Why or why not?
2. Find a spanning set for $\text{Nul } A$ where $A = \begin{bmatrix} 1 & 3 & 5 & 0 \\ 0 & 1 & 4 & -2 \end{bmatrix}$.
3. Find A such that the given set is $\text{Col } A$:

$$\left\{ \begin{bmatrix} b - c \\ 2b + c + d \\ 5c - 4d \\ d \end{bmatrix} : b, c, d \text{ real} \right\}.$$

4. Let $T : V \rightarrow W$ be a linear transformation from a vector space V into a vector space W . Prove that $\text{Ker } T$ is a subspace of V .

Answers or Hints are on the back.

Answers or Hints:

#1: Yes for both Nul A and Col A . Explain why. Since $Ax = 0$, x is an eigenvector corresponding to eigenvalue what?, So the answer is Yes again.

$$\#2: \left\{ \begin{bmatrix} 7 \\ -4 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -6 \\ 2 \\ 0 \\ 1 \end{bmatrix} \right\}$$

$$\#3: \begin{bmatrix} 1 & -1 & 0 \\ 2 & 1 & 1 \\ 0 & 5 & -4 \\ 0 & 0 & 1 \end{bmatrix}$$

#4: By definition, $\text{Ker } T$ is a subset of V . Now, show three other items. For example, if $u \in \text{Ker } T$ and $v \in \text{Ker } T$, is $u + v \in \text{Ker } T$? Why?